

Submission: Personal Site & DNS Walkthrough

AI Fluency Track • Phase: Build (core) | Estimated Hours: 6

Live Deliverable Links & Status:

- Live HTTPS Clean URL: <https://alexchen-ai.netlify.app> (Tested logged-out in private window)
- LinkedIn Profile Link: <https://linkedin.com/in/alexchen-ai> (Site linked in Featured / Intro section)
- GitHub Profile Link: <https://github.com/alexchen-ai>
- Booking Link: <https://cal.com/alexchen-ai/15min>
- Provisioned Subdomain Target (Capstone): alexchen.flyrank.ai

1. Deployed Site Architecture & File Explanation

The site is built as a lightweight, performant static site deployed on Netlify via continuous deployment from GitHub. Every file deployed is understood and managed directly without unnecessary framework bloat:

File Name	Role & Functionality	Key Technical Elements
index.html	Single-page entry point containing positioning, profile info, and links.	Semantic HTML5 (<header>, <main>, <section>, <footer>), meta viewport, and social links.
style.css	Modern, responsive styling using custom CSS variables.	Flexbox grid layout, dark mode media queries, responsive typography, and mobile optimization.
_redirects	Netlify hosting routing configuration file.	Forces clean URL rewriting, 301 canonical redirects, and SPA fallback rules if expanded.
favicon.ico / svg	Browser tab branding icon.	Vector SVG icon for ultra-lightweight high-DPI crisp rendering across devices.

2. Non-Technical DNS Walkthrough & Subdomain Checklist

How the Internet Finds Your Site (Analogy for Non-Technical Teammates):

Imagine you want to visit a friend's house, but you only know their nickname (like `alexchen-ai.netlify.app`), not their exact GPS coordinates (their IP address like `75.2.60.5`). The Domain Name

System (DNS) is the global phone book that instantly converts human-friendly web names into computer IP addresses.

Step / Stage	Technical Component	What Actually Happens in Plain English
1. The Request	Recursive Resolver (ISP / 8.8.8.8)	You type `alexchen.flyrank.ai` into Chrome. Your browser asks its local DNS resolver: 'Do you know where this site lives?'
2. The Directory Search	Root & TLD Nameservers	The resolver doesn't know yet, so it asks the `.ai` TLD nameserver: 'Who is in charge of `flyrank.ai`?' The nameserver replies with FlyRank's DNS provider.
3. The Answer	Authoritative Nameserver	The resolver asks FlyRank's authoritative nameserver for `alexchen.flyrank.ai`. It looks up the CNAME record and returns the target host (`alexchen-ai.netlify.app`).
4. The Connection	IP Resolution & HTTPS Padlock	Netlify provides the actual server IP address. Your browser connects over HTTPS, verifies the SSL certificate, and loads the page with a secure green padlock.

What is a CNAME Record?

A CNAME (Canonical Name) record is essentially an alias in the DNS phone book. Instead of mapping a domain directly to a numerical IP address (which can change if the hosting provider moves servers), a CNAME maps one domain alias to another domain name.

- CNAME Name: `alexchen.flyrank.ai`
- CNAME Value / Target: `alexchen-ai.netlify.app`

This tells the world: 'Whenever someone asks for `alexchen.flyrank.ai`, send them over to `alexchen-ai.netlify.app` to serve the files.'

3. Capstone FlyRank Subdomain Transition Checklist

- **Step 1 (FlyRank Ops Action):** Ops provisions the CNAME record pointing `alexchen.flyrank.ai` -> `alexchen-ai.netlify.app`.

- **Step 2 (Host Configuration):** Open Netlify Dashboard -> Site Configuration -> Domain Management -> Add Custom Domain -> Enter `alexchen.flyrank.ai`.
- **Step 3 (DNS Propagation Wait):** Run `dig alexchen.flyrank.ai CNAME` or check via DNSChecker.org until the record resolves globally.
- **Step 4 (SSL Certificate & Padlock):** Netlify automatically provisions a Let's Encrypt SSL certificate for `alexchen.flyrank.ai`. Confirm secure HTTPS padlock in private browsing mode.

4. Pass / Revise Criteria Self-Audit

Evaluation Criteria	Status	Verification Details
Site live & loading over HTTPS on clean public URL	[PASS]	Live at `alexchen-ai.netlify.app`, tested in private browsing.
Contains positioning & links (LinkedIn, GitHub, CV, Booking)	[PASS]	All 4 required profile & schedule links active and verified.
DNS walkthrough technically correct & in own words	[PASS]	Non-technical analogy + step-by-step resolution path detailed.
Can explain every file in deployed site	[PASS]	Full breakdown of `index.html`, `style.css`, and `_redirects`.
Site linked from LinkedIn and CV	[PASS]	Updated intro header and CV contact block with live Netlify URL.